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The picokelvin temperature range will be approached by evaporative cooling followed by adiabatic expansion which lowers the energy even further. Although the theoretical limits of evaporative cooling are in the picokelvin range, the speed of cooling becomes extremely slow because the cooling rate is proportional to the rate of elastic collisions and therefore to the thermal velocity. However, once a Bose condensate has been created at the current critical temperatures of 100 nK to 1 microK, one might adiabatically expand the condensate and thus obtain very dilute condensates with much reduced zero-point energy and interaction energy. The major challenge is that the field curvatures have to be reduced to 10^-5 G/cm^2 while a magnetic field gradient of 8 G/cm is required to compensate for gravity in the laboratory. Such expanded Bose condensates would be coherent matter waves with de Broglie wavelengths approaching 1 mm which are not only very promising for precision metrology, but also for atom optics and conceptional experiments in quantum mechanics.

Task Description:	A systematic study of the limiting processes in generating such pK samples of atoms will be performed. Possible limitations are the precise control of the magnetic fields and heating due to background gas collisions and vibrations. We expect that these processes will become more critical at lower temperatures. Precise RF (or Raman) spectroscopy can be used to measure fundamental constants like h/m and to test fundamental symmetries like charge neutrality and time reversal. RF spectroscopy of Bose condensates will be performed, both in free fall and confined in a magnetic trap. The first steps will focus on the study of systematic shifts due to the trapping potential and the presence of the Bose condensate at finite density. results from this study will naturally lead to the formulation of a concept how to implement RF spectroscopy with mHz linewidth in the microgravity environment of a space station. The goal is to produce atomic samples with picokelvin temperatures and to improve precision spectroscopy. In microgravity, picokelvin samples would spread out by less than 1 cm/minute and would allow rf and microwave spectroscopy with mHz linewidths, an improvement of three orders of magnitude over earth-based atomic fountain experiments. RF spectroscopy of such ultracold samples will be performed to study shifts and broadening due to the finite atomic density and the presence of the trapping potential. This study should lead to a clear concept of how to perform precision spectroscopy with mHz linewidth on a space station. This holds the promise of greatly improving current atomic clocks and experiments testing fundamental symmetries in physics. This task develops key technologies for preparing picokelvin samples of atoms for precision spectroscopy. At this point, the major limitation for spectroscopy will be gravity. Picokelvin atoms in a microgravity environment holds the promise of greatly improving current atomic clocks and experiments testing fundamental symmetries in physics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. Collisions in zero temperature Fermi gases The smoking gun of Bose-Einstein condensation has been the anisotropic “superfluid” expansion of elongated condensates released from the trap. Similarly, superfluid Fermi gas would show anisotropic expansion due to superfluid hydrodynamics [1]. A recent observation of anisotropic expansion of an ultracold, interacting, two-spin fermionic mixture [2] has created considerable excitement and raised the question under what conditions is this expansion a signature of fermionic superfluidity and not of collisional hydrodynamics. We examined the collisional behavior of two-component Fermi gases released at zero temperature from a harmonic trap [3]. Using a phase-space formalism to calculate the collision rate during expansion, we find that Pauli blocking plays only a minor role for momentum changing collisions. As a result, for a large scattering cross-section, Pauli blocking will not prevent the gas from entering the collisionally hydrodynamic regime. In contrast to the bosonic case, hydrodynamic expansion at very low temperatures is therefore not evidence for fermionic superfluidity. 2. Spectroscopic insensitivity to cold collisions in a two-state mixture of fermions We have experimentally addressed the relation between coherence and spectroscopic measurements in a binary mixture of ultracold fermions. We demonstrated that shifts of spectroscopic lines are absent even in a fully decohered binary mixture, in which the particles are distinguishable, and the many-body mean-field energy in the system has developed [4]. We theoretically showed that this is a direct consequence of the coherent nature of the RF excitation, and is not dependent on the coherence of the sample on which spectroscopy is performed. Our calculation intuitively explains both our results for fermions, and previous results for bosons obtained in Boulder [5]. 3. Cooling of Bose-Einstein condensates below 500 Picokelvin The lowest temperatures for trapped atoms are usually achieved in low-density samples. At high densities, interaction effects adversely affect the cooling process and the temperature diagnostics. We have achieved a new record-low temperature of less than 500 picokelvin in a very weak trap using a combination of gravitational and magnetic forces [6]. The partially condensed atomic vapors were adiabatically decompressed by weakening the gravito-magnetic trap to a mean frequency of 1 Hertz, then evaporatively reduced in size to 2500 atoms. This lowered the peak condensate density to 5 x 1010 atoms per cubic centimeter and cooled the entire cloud in all three dimensions to a kinetic temperature of 450 ± 80 picokelvin. These samples are characterized by a thermal velocity of 1 mm/s, a speed of sound of 100 mm/s, and a healing length limited by the 20 mm harmonic oscillator length of the trapping potential. Low temperature and low-density ensembles are important for spectroscopy, metrology, and atom optics. In addition, they are predicted to experience quantum reflection from material surfaces. 4. Formation of Quantum-Degenerate Sodium Molecules A current frontier in the field of ultracold gases is the study of ultracold molecules. In 2003, several groups succeeded in converting ultracold atoms into ultracold molecules by magnetically tuning a molecular level close to zero binding energy (Feshbach resonance). Atoms can then form molecules without release of heat. In our experiment, we produced ultracold sodium molecules from an atomic Bose-Einstein condensate by ramping an applied magnetic field across a Feshbach resonance [7]. More than 105 molecules were generated with a conversion efficiency of ~4%. High phase-space density could only be achieved by rapidly removing residual atoms, before atom-molecule collisions caused trap loss and heating. This was accomplished by a new technique for preparing pure molecular clouds, where light resonant with an atomic transition selectively ‘‘blasted’’ unpaired atoms from the trap. Time-of-flight analysis of the pure molecular sample yielded an instantaneous phase-space density greater than 20. 5. Distillation of Bose-Einstein condensates in a double-well potential The characteristic feature of Bose-Einstein condensation is the accumulation of a macroscopic number of particles in the lowest quantum state. Condensate fragmentation, the macroscopic occupation of two or more quantum states, is usually prevented by interactions [8]. However, multiple condensates may exist in metastable situations. Let's assume that an equilibrium condensate has formed in one quantum state, but now we modify the system allowing for one even lower state. How does the original condensate realize that it is in the wrong state and eventually migrate to the true ground state of the system? What determines the time scale for this equilibration process? This is the situation, which we have experimentally explored by preparing a Bose-Einstein condensate in an optical dipole trap and distilling it into a second empty dipole trap adjacent to the first one [9]. The distillation was driven by thermal atoms spilling over the potential barrier separating the two wells and then forming a new condensate. This process serves as a model system for metastability in condensates, provides a test for quantum kinetic theories of condensate formation, and also represents a novel technique for creating or replenishing condensates in new locations. 6. Dissociation and Decay of Ultracold Sodium Molecules We have studied the dissociation and decay of ultracold molecules. Sodium molecules were formed in a highly excited vibrational state by recombining two ultracold atoms [9]. An external magnetic field “tuned” the molecular binding energy close to zero (Feshbach resonance) allowing resonant

recombination. By ramping up the magnetic field, the molecular level was moved into the continuum, and the molecule dissociated. When the magnetic field ramp is very slow, the molecules follow adiabatically and end up in the lowest energy state of the atoms. The dissociation products will populate higher-lying atomic states if the ramp is fast (compared to the strength of the coupling between the molecular and atomic states). Therefore, from the observed dissociation energies, the strength of the atom-molecule coupling could be determined. The non-linear dependence of the dissociation energy on the ramp speed reflects the Wigner threshold law for the onset of dissociation: The dissociation lifetime decreases when the molecular energy is higher above threshold. Furthermore, inelastic molecule-molecule and molecule-atom collisions were characterized. The rapid inelastic decay imposes a severe limit to further evaporative cooling.

7. Observation of Bose-Einstein Condensation of Molecules Molecular condensates could lead to a host of new scientific explorations. These include quantum gases with anisotropic dipolar interactions, tests of fundamental symmetries such as the search for a permanent electric dipole moment, study of rotational and vibrational energy transfer processes, and coherent chemistry, where reactants and products are in coherent quantum superposition states. In November 2003, observation or indirect evidence for Bose-Einstein condensation of molecules was reported by three groups [10-12]. We observed BEC of lithium molecules. When a spin mixture of fermionic 6Li atoms was evaporatively cooled in an optical dipole trap near a Feshbach resonance, the atomic gas was converted into 6Li molecules. Below 600 nK, a Bose-Einstein condensate of up to 900,000 molecules was identified by the sudden onset of a bimodal density distribution. This condensate realizes the limit of tightly bound fermion pairs in the crossover between BCS superfluidity and Bose-Einstein condensation.

8. Condensation of Pairs of Fermionic Atoms near a Feshbach Resonance Ultracold atomic gases have become a medium to realize novel phenomena in condensed matter physics and test many-body theories in new regimes. Of particular interest are pairing phenomena in fermionic gases, which have direct analogies to superconductivity [13]. The interactions which drive the pairing in these gases can be controlled using a Feshbach resonance. This provides an opportunity to experimentally probe what is known as the BCS-BEC crossover; as the strength of the effective attractive interaction between particles is increased a continuous transition from condensation of delocalized Cooper pairs to condensation of tightly bound bosonic molecules is predicted. We have observed Bose-Einstein condensation of pairs of fermionic atoms in an ultracold 6Li gas at magnetic fields above a Feshbach resonance, where no stable 6Li2 molecules would exist in vacuum [14]. Similar to Ref.. [15], those pairs were observed after rapidly transferring them to stable molecules by sweeping the magnetic field across the Feshbach resonance, and then detecting the molecular signal. Condensate fractions as high as 80% were obtained. The large condensate fractions are interpreted in terms of preexisting molecules, which are quasi-stable even above the two-body Feshbach resonance due to the presence of the degenerate Fermi gas.

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Bibliography Type:	Description: (Last Updated: 02/05/2006)
Articles in Peer-reviewed Journals	M.W. Zwierlein, C.A. Stan, C.H. Schunck, S.M.F. Raupach, A.J. Kerman, and W. Ketterle "Condensation of Pairs of Fermionic Atoms Near a Feshbach Resonance" Phys. Rev. Lett. 92, 120403 (2004) , Jan-2004
Articles in Peer-reviewed Journals	T. Mukaiyama, J. R. Abo-Shaeer, K. Xu, J. K. Chin, and W. Ketterle "Dissociation and Decay of Ultracold Sodium Molecules" Phys. Rev. Lett. 92, 180402 (2004) , Jan-2004
Articles in Peer-reviewed Journals	Y. Shin, M. Saba, A. Schirotzek, T.A. Pasquini, A.E. Leanhardt, D.E. Pritchard, and W. Ketterle "Distillation of Bose-Einstein condensates in a double-well potential" Phys. Rev. Lett. 92, 150401 (2004) , Jan-2004
Articles in Peer-reviewed Journals	S. Gupta, Z. Hadzibabic, J.R. Anglin, and W. Ketterle "Collisions in zero temperature Fermi gases" Phys. Rev. Lett. 92, 100401 (2004) , Jan-2004
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Articles in Peer-reviewed Journals	K. Xu, T. Mukaiyama, J.R. Abo-Shaeer, J.K. Chin, D. Miller, and W. Ketterle "Formation of Quantum-Degenerate Sodium Molecules" Phys. Rev. Lett. 91, 210402 (2003) , Jan-2004
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